RNA-seq Analysis

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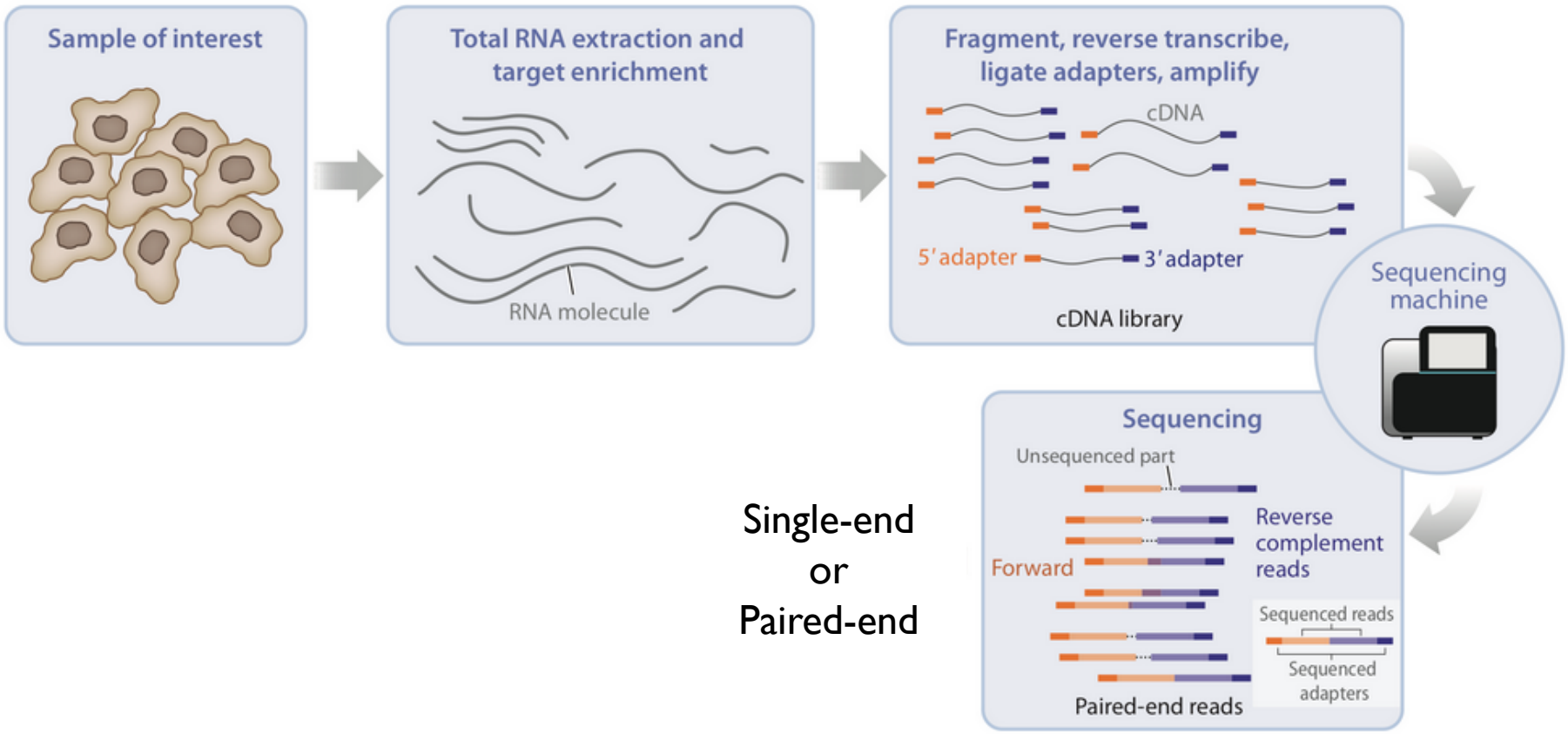
RNA-seq is used for…

- Quantifying the abundance of transcripts
- Identify and quantify expression changes among conditions
- Splicing analysis
- Non-coding RNA quantification
- SNVs detection
- Transcriptome assembly
- …

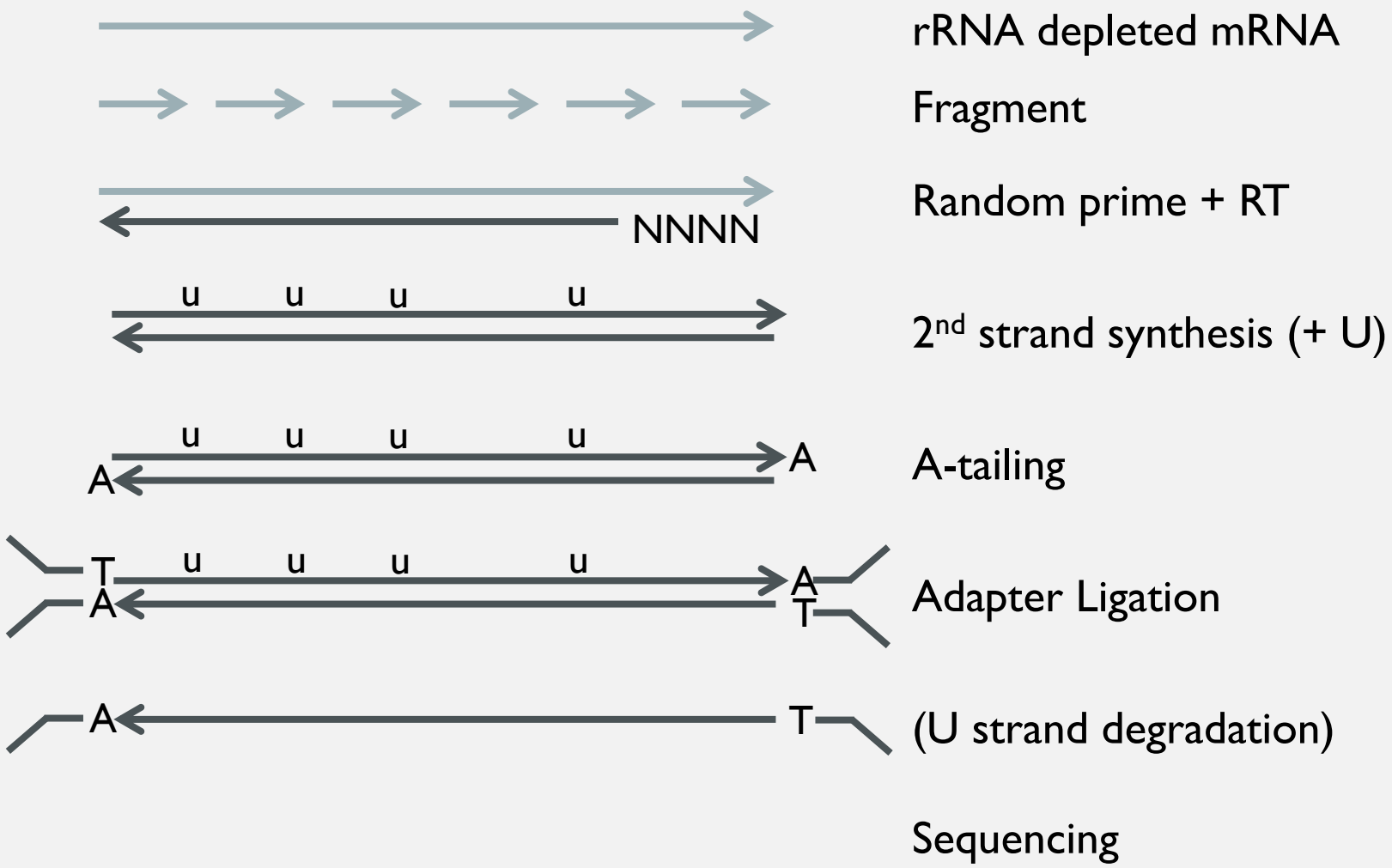
RNA-seq protocol



Single-end
or
Paired-end

Figure: The sequencing workflow. Image adapted from Van den Berge et al. (2019).

Stranded RNA-seq



Introduction to Linux

Terminal layout

```
user@server:~$ ls
01_reads  02_reference  03_hisat2  04_featureCounts  05_deseq2  06_GSEA
cheat_sheet.md  deseq2_env.yml  rnaseq_env.yml

user@server:~$
```

- Command prompt - you can't enter a command unless you can see this
- The command we're going to run (`ls` in this case, to list files)
- The output of the command - just text in this case

Structure of Unix commands

Comments that are not executed

command --options (-o) arguments

- *Command/Program name*: instruction to be executed.
- *Options/Switches*: modifiers of command behaviour.
- *Arguments/Data*: objects on which the command acts.

Example:

```
# Show the first 10 lines of a file.
head my_text.txt

# Show the first 100 lines of a file.
head --lines 100 my_text.txt

# Show the first 100 line of a file
head -n 100 my_text.txt
```

Specifying file paths

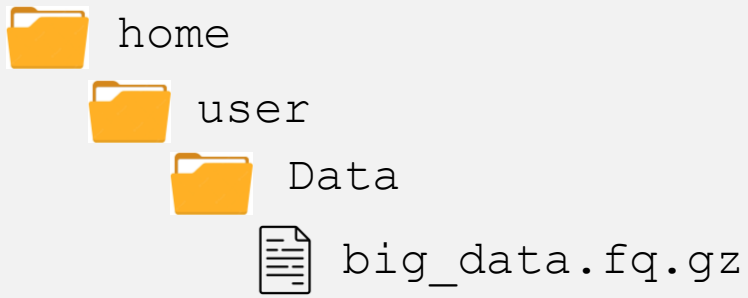
If I'm in `/home/user`

- Absolute path:

  `/home/user/Data/big_data.fq.gz`

- Relative path:

  `Data/big_data.fq.gz`



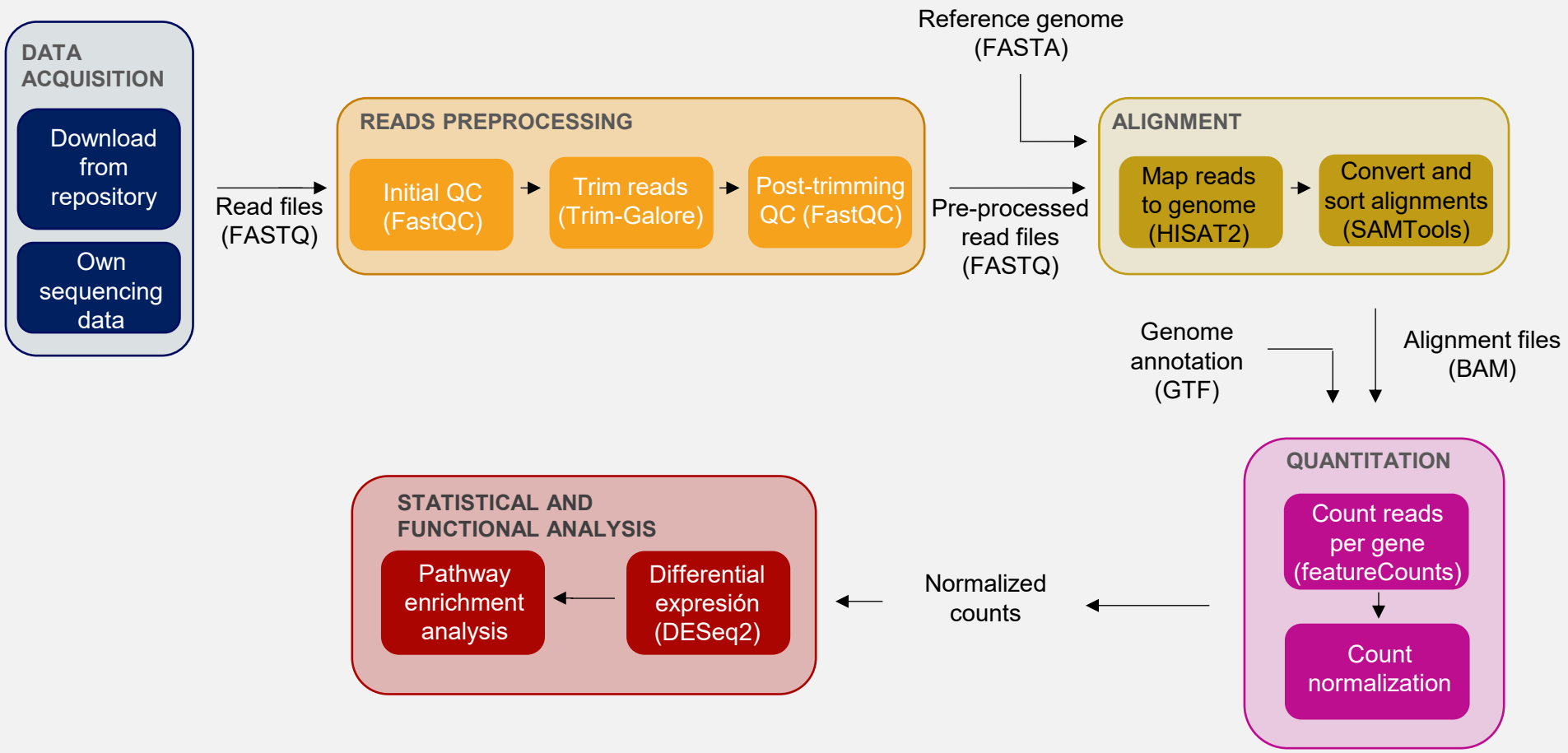
Command line completion

- Command line completion is achieved by typing a partial path and then pressing the TAB key (to the left of Q).

- You should use TAB completion to fill in paths for locations which exist so you can't make typing mistakes.

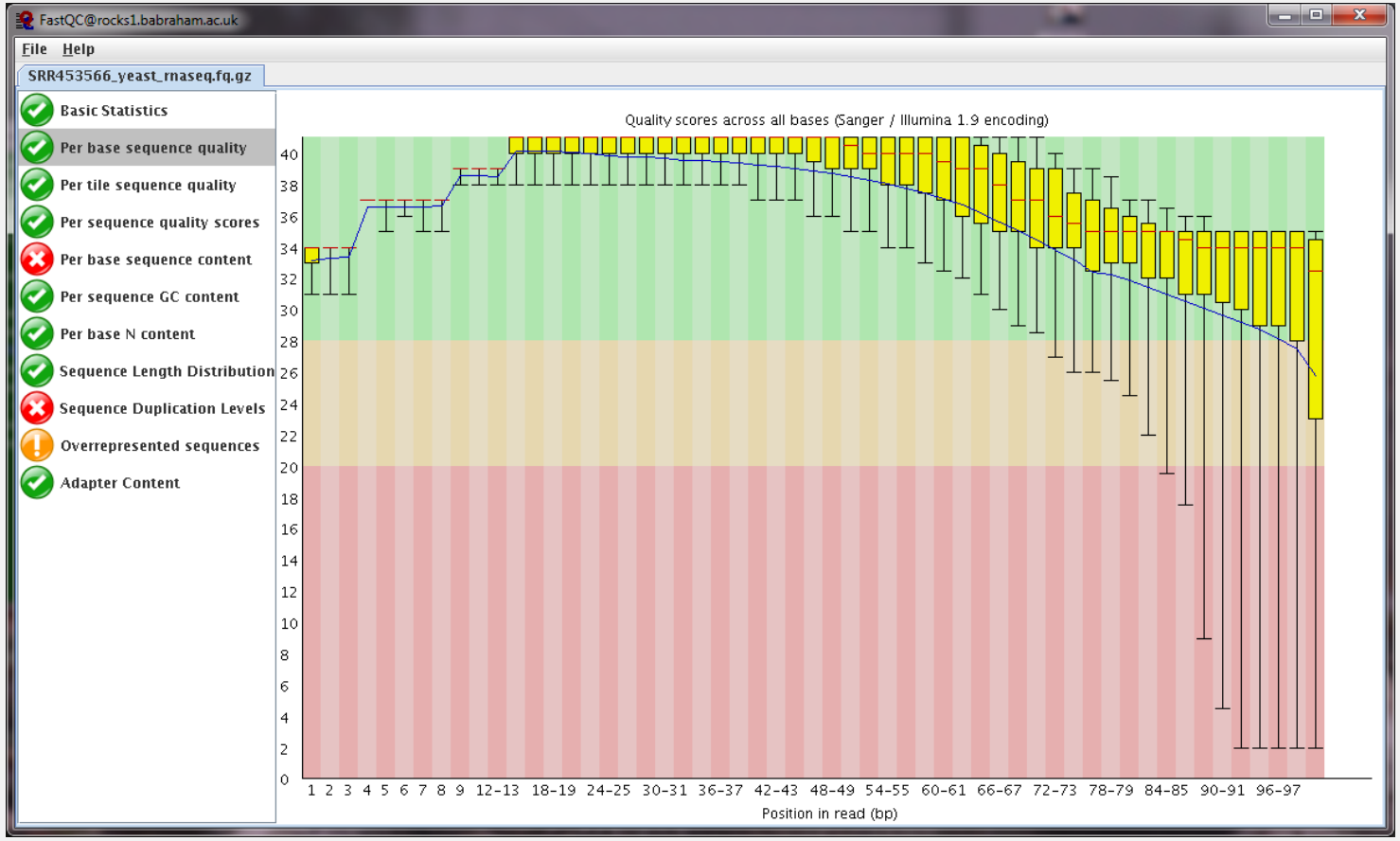
RNA-seq analysis

Analysis flowchart



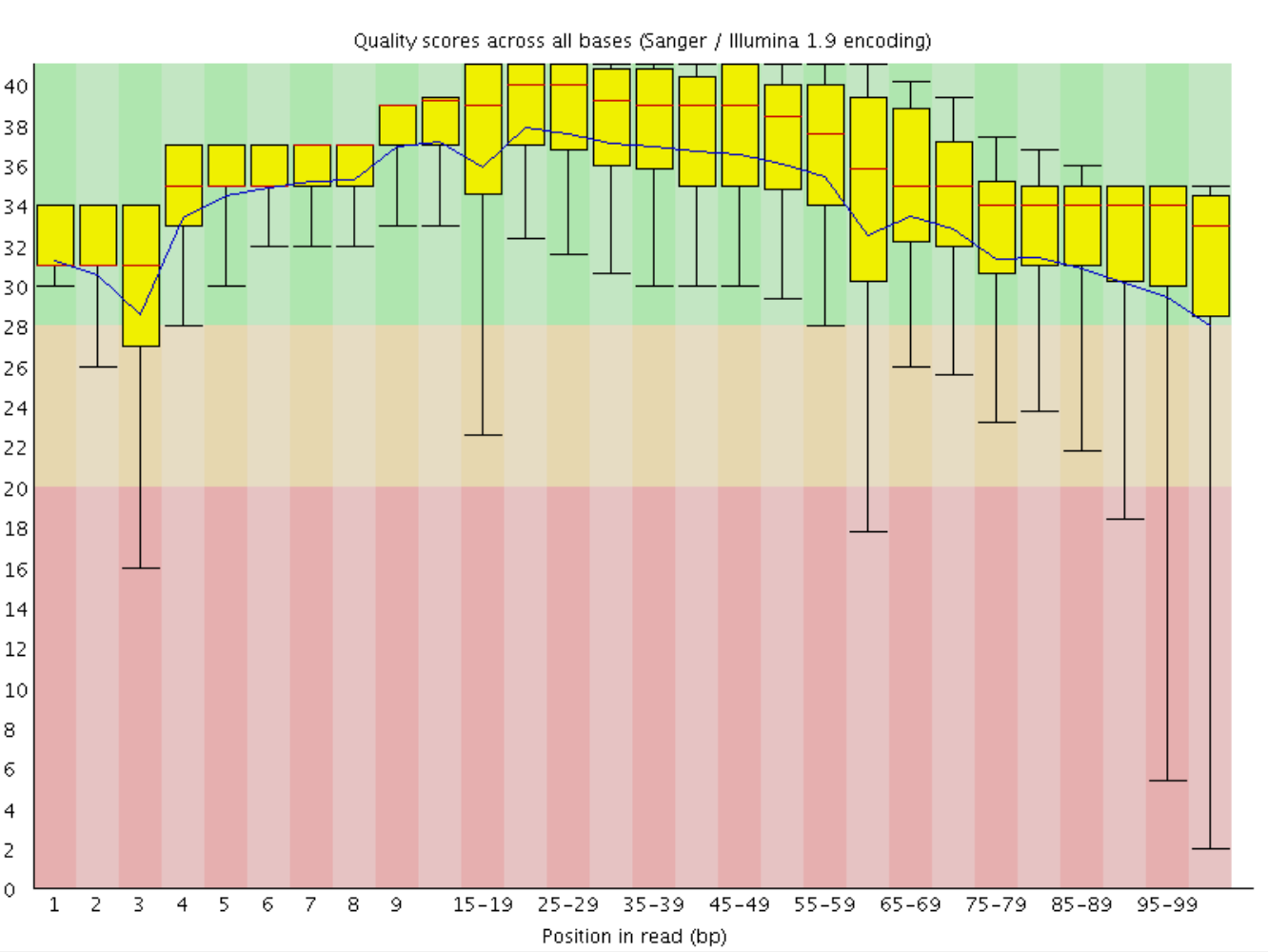
1. Raw sequence quality control (QC)

- Base call quality
- Composition
- Duplication
- Contamination



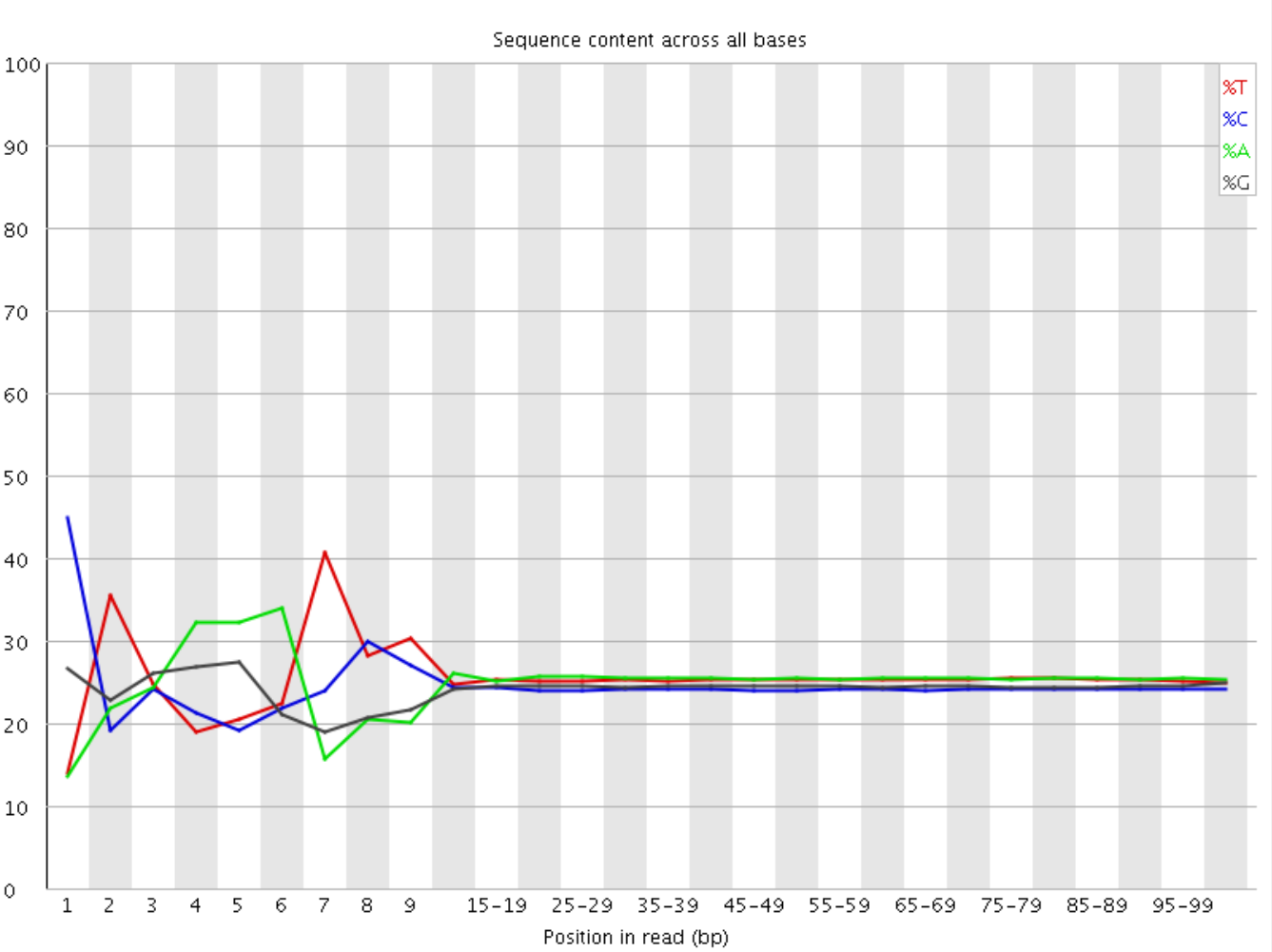
1. Raw sequence quality control (QC)

Call Quality
(Phred score)



Read Position

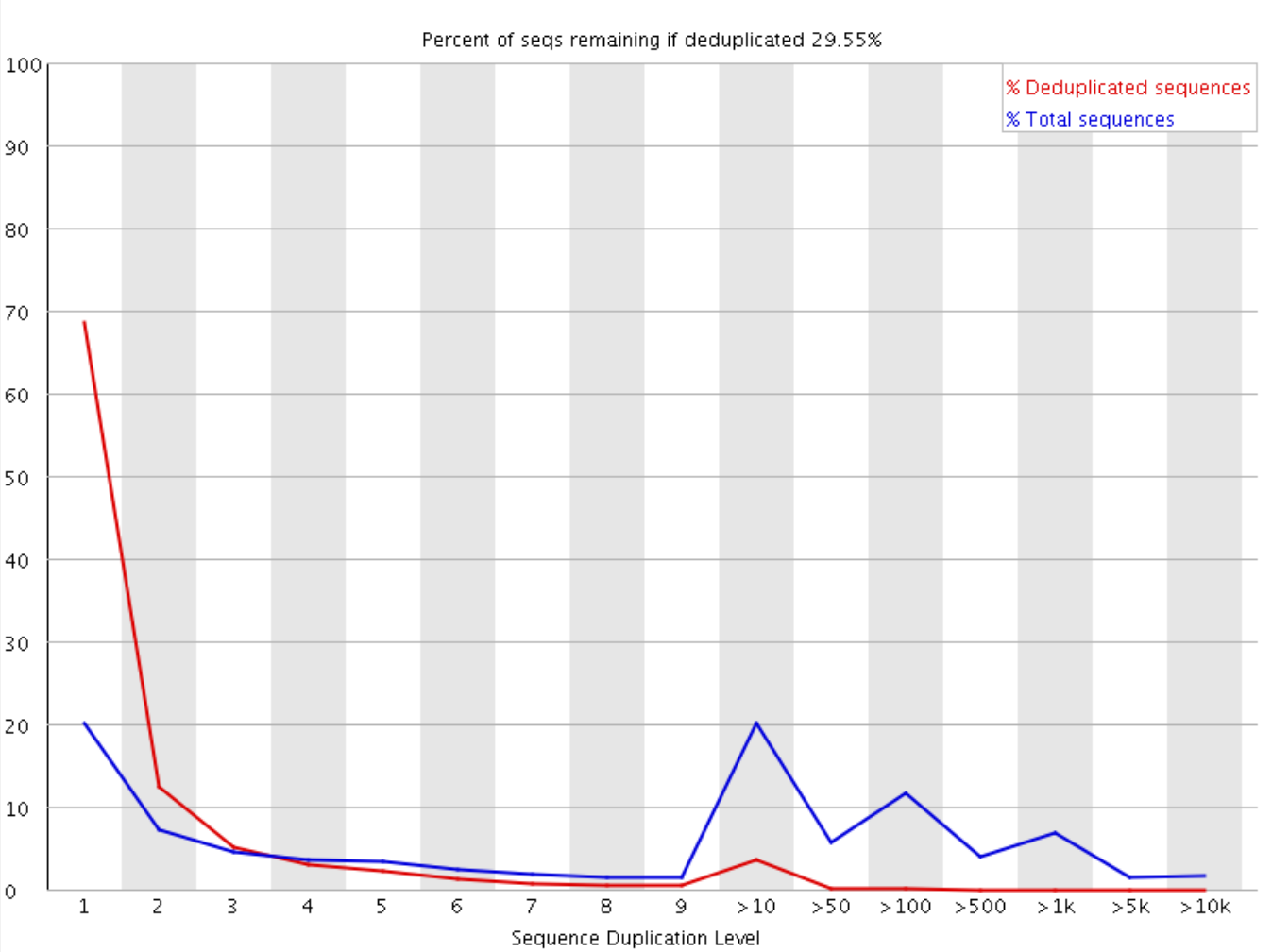
1. Raw sequence quality control (QC)



Read Position

1. Raw sequence quality control (QC)

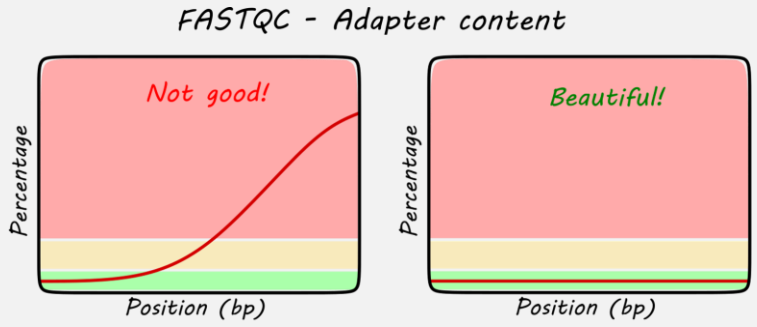
Percentage of library

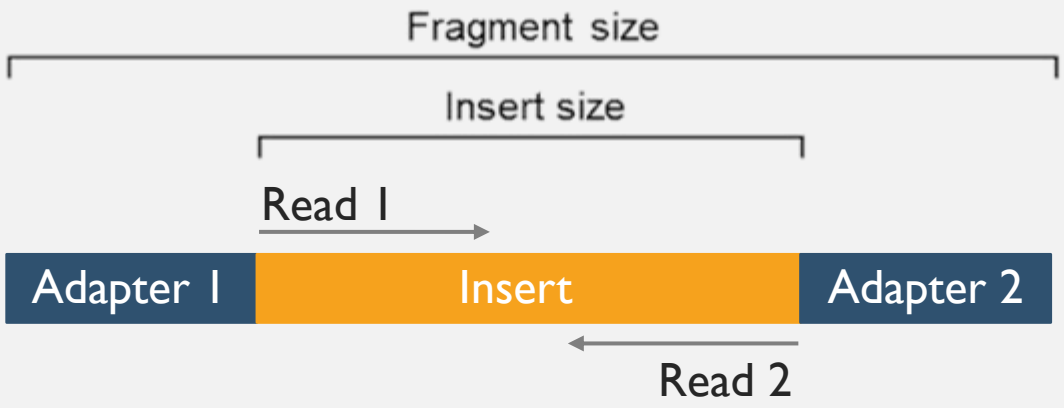


Level of duplication

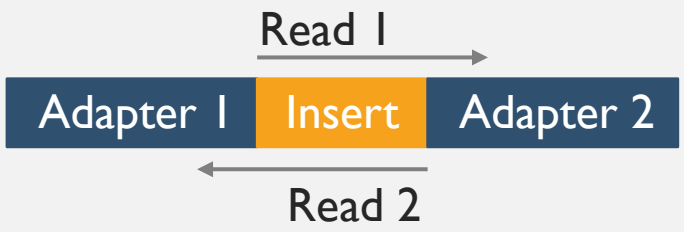
2. Trimming

- Removes adapter sequences
- Remove very short sequences (default: < 20bp)



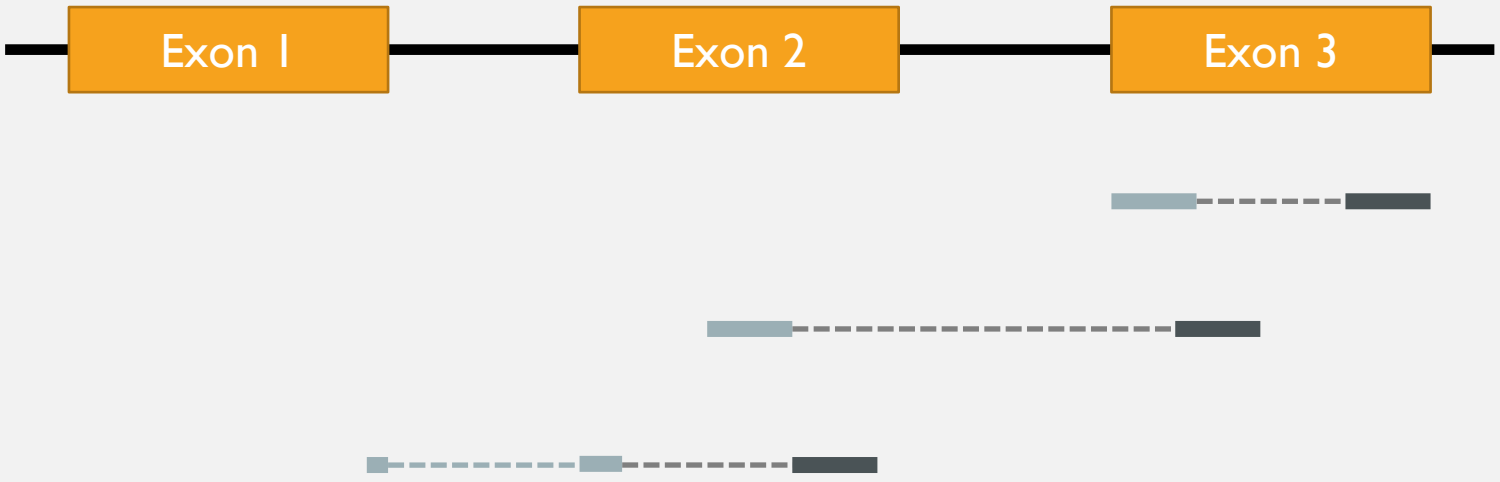


Reads without adapters



Reads **WITH** adapters

3. Alignment



Genome

Simple mapping within exons

Mapping between exons

Spliced mapping

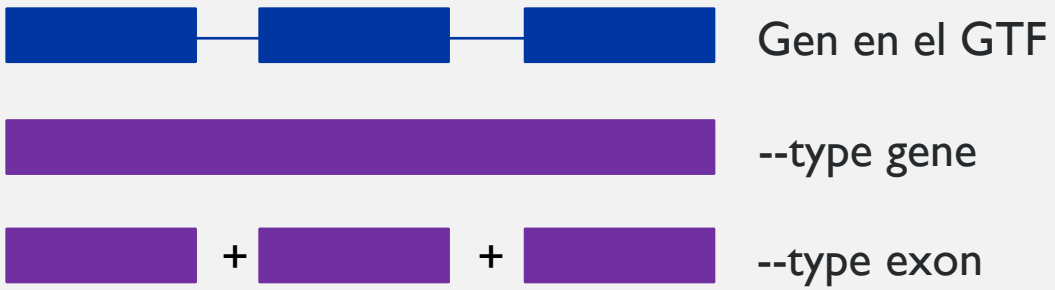
Discarded by non-splice-aware software

Splice-aware software:

- HISAT2
- STAR
- …

4. Reads per gene quantification

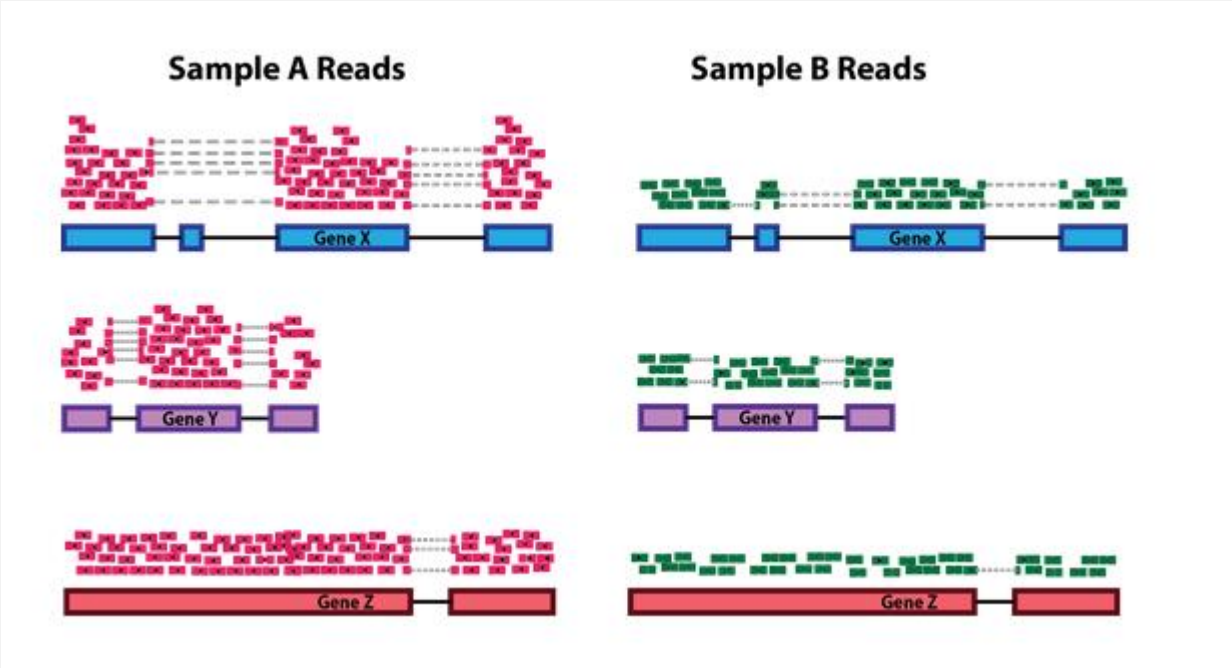
Counting the number of reads per gene (sum of reads per exons)



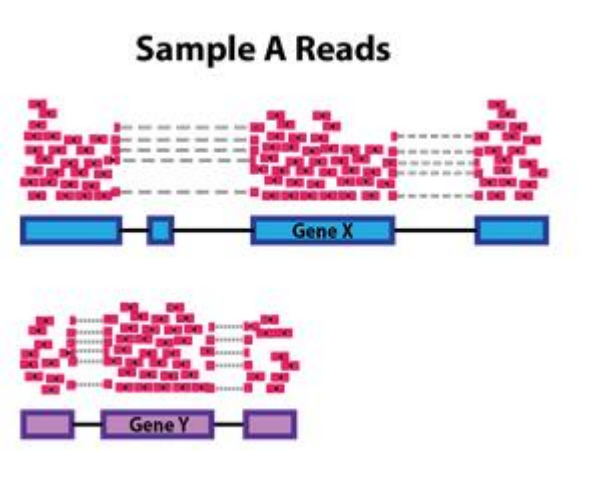
5. Differential expression

5.1. Read counts normalization

Sequencing depth



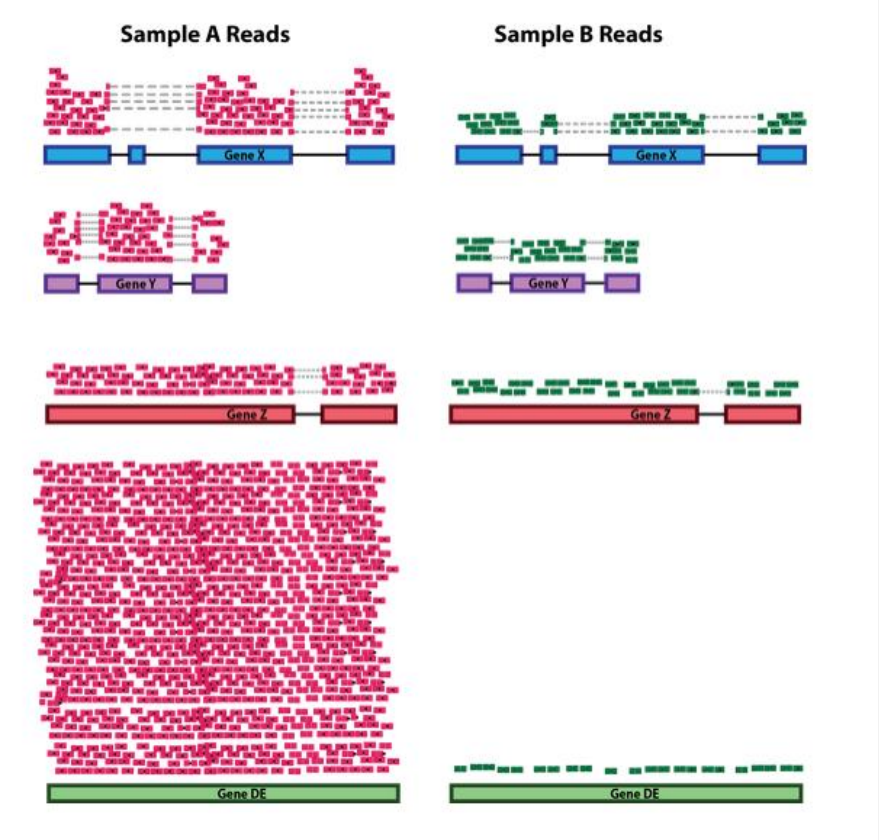
Gene length

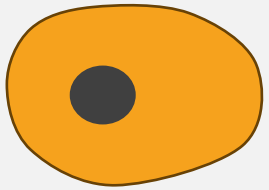


5. Differential expression

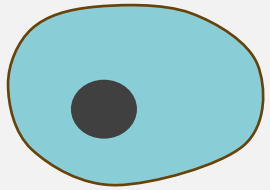
5.1. Read counts normalization

RNA composition





Orange cell



Blue cell

Gene A:	10 transcripts (10%)	10 transcripts (50%)
Gene B:	10 transcripts (10%)	10 transcripts (50%)
Gene C:	80 transcripts (80%)	no expressed

We sequence the two cells with 10 reads
(same number of total reads)

Gene A and B are expressed 5 more times in
the blue cell

5. Differential expression

5.1. Read counts normalization

Common normalization methods

Several common normalization methods exist to account for these differences:

Normalization method	Description	Accounted factors	Recommendations for use
CPM (counts per million)	counts scaled by total number of reads	sequencing depth	gene count comparisons between replicates of the same samplegroup; **NOT for within sample comparisons or DE analysis**
TPM (transcripts per kilobase million)	counts per length of transcript (kb) per million reads mapped	sequencing depth and gene length	gene count comparisons within a sample or between samples of the same sample group; **NOT for DE analysis**
RPKM/FPKM (reads/fragments per kilobase of exon per million reads/fragments mapped)	similar to TPM	sequencing depth and gene length	gene count comparisons between genes within a sample; **NOT for between sample comparisons or DE analysis**
DESeq2's **median of ratios** [1]	counts divided by sample-specific size factors determined by median ratio of gene counts relative to geometric mean per gene	sequencing depth and RNA composition	gene count comparisons between samples and for **DE analysis**; **NOT for within sample comparisons**
EdgeR's **trimmed mean of M values (TMM)** [2]	uses a weighted trimmed mean of the log expression ratios between samples	sequencing depth, RNA composition, and gene length	gene count comparisons between and within samples and for **DE analysis**

6. Pathway enrichment analysis

Over-representation análisis (ORA)

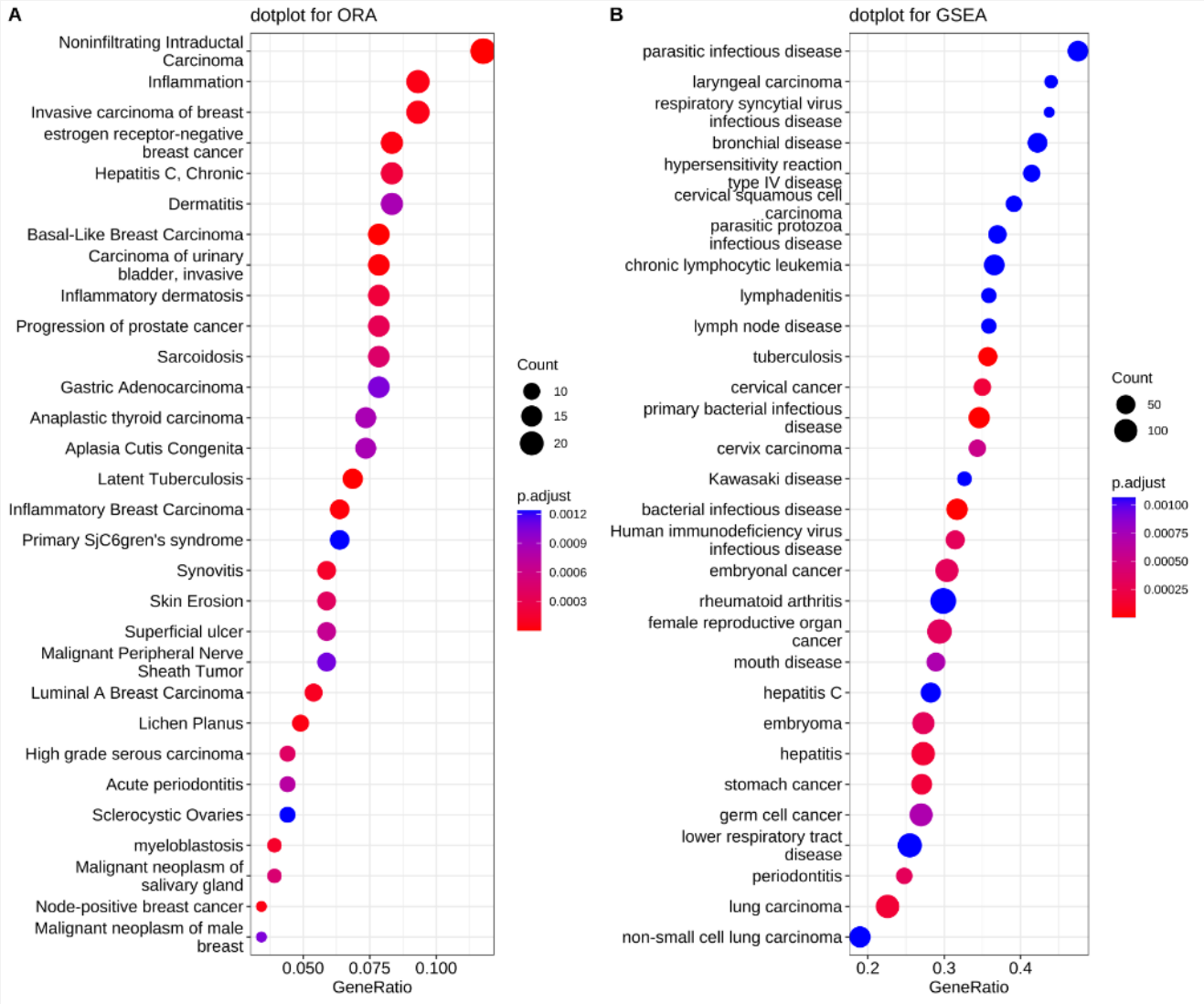
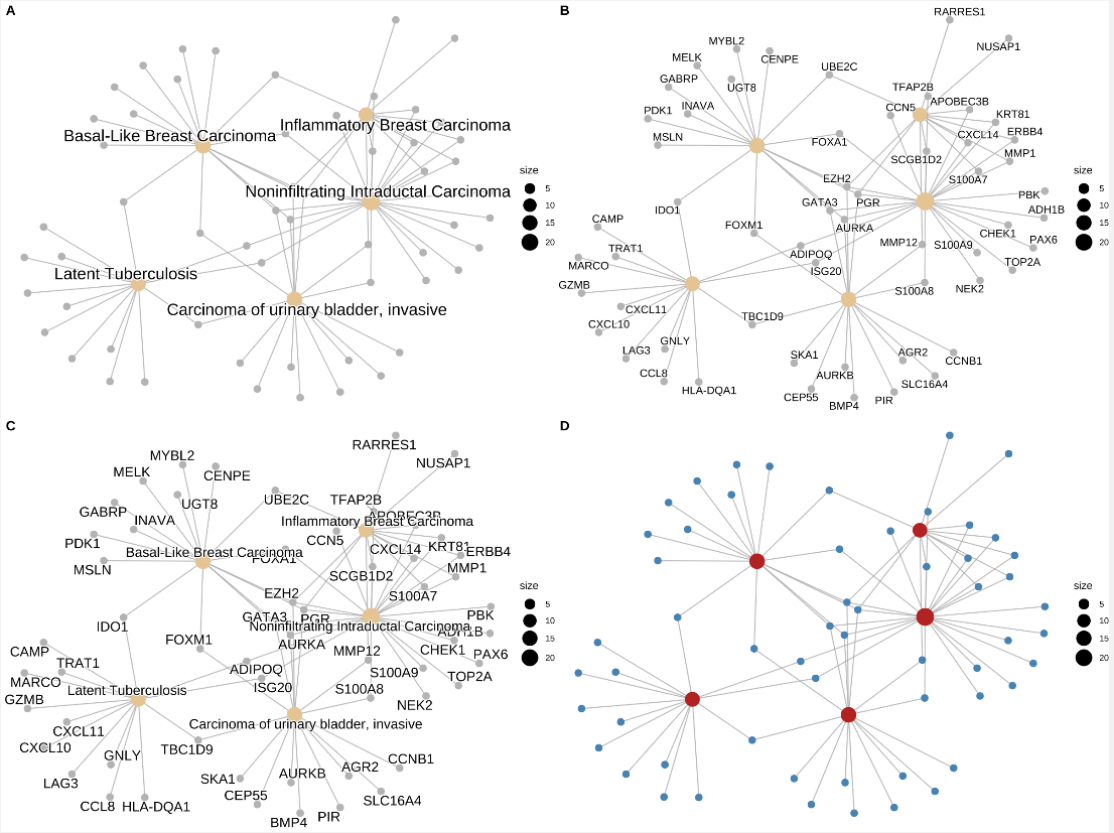
Evalúa si ciertos conjuntos de genes (gene sets, ontologías, rutas) están sobrerrepresentados en tu lista comparado con un background de referencia (todas los genes posibles).

Input: lista de genes asociados a DARs (por ejemplo, genes cercanos a regiones accesibles diferencialmente).
Output: listado de ontologías o rutas enriquecidas, sin considerar dirección o magnitud de cambio.

Gene set enrichment análisis (GSEA)

Ordena todos los genes por su magnitud de cambio y evalúa si los genes de un conjunto específico tienden a aparecer al inicio o al final de la lista.

Input: lista de genes asociados a DARs con valores cuantitativos (p. ej., log2 fold change de accesibilidad o expresión).
Output: listado de rutas o ontologías enriquecidas en genes up- o down-regulados, considerando dirección y magnitud del cambio.

http://geneontology.org/ https://www.genome.jp/kegg/ https://reactome.org/

6. Pathway enrichment analysis